

No. of Functional Features Included in the Solution

Date	7 July 2026
Team ID	SWTID-2026-3702
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Module / Component	Feature Description	Status	Marks Contribution
1	Dataset Loading & Inspection	Epic 3 — Jupyter Notebook	Load HDI.csv with 195 rows x 82 cols using pd.read_csv(), display head() and unique() country values	Done	✓
2	Exploratory Data Analysis	Epic 3 — Jupyter Notebook	Generate strip plots (Life Exp vs HDI, Schooling vs HDI) and correlation heatmap using Seaborn	Done	✓
3	Feature Selection	Epic 4 — Jupyter Notebook	Select X (Country, Life Exp, Schooling, GNI, Internet) and y (HDI Score) using iloc[]	Done	✓
4	Null Value Handling	Epic 4 — Jupyter Notebook	Detect null values with isnull().sum() and fill using column mean (fillna(X.mean()))	Done	✓
5	Label Encoding	Epic 4 — Jupyter Notebook	Encode Country column from string to numeric using sklearn LabelEncoder	Done	✓
6	Train/Test Split	Epic 5 — Jupyter Notebook	Split dataset 90% train (175 rows) / 10% test (20 rows) with random_state=42	Done	✓
7	Linear Regression Model	Epic 6 — Jupyter Notebook	Train LinearRegression model on x_train/y_train, predict on x_test, R-squared = 93.74%	Done	✓
8	Model Evaluation & Visualization	Epic 6 — Jupyter Notebook	Calculate R-squared score, plot Actual vs Predicted scatter chart	Done	✓
9	Model Serialization	Epic 7 — model/HDI.pkl	Save trained model using pickle.dump() to HDI.pkl (732 bytes)	Done	✓

10	Flask Home Page	Epic 8 — home.html	Landing page with HDI description, navigation bar, 'Get Prediction' button	Done	✓
11	Prediction Input Form	Epic 8 — indexnew.html	Country dropdown + 4 input fields with Flask POST form to /predict route	Done	✓
12	HDI Prediction Engine	Epic 8 — app.py predict()	Load HDI.pkl, accept user inputs, predict score, classify into HDI category	Done	✓
13	Result Display Page	Epic 8 — resultnew.html	Show predicted HDI score and category (Very High/High/Medium/Low) with styling	Done	✓
14	Flask Navigation	Epic 8 — app.py	3 routes: / (home), /Prediction (form), /predict (POST) for full navigation	Done	✓

Feature Summary:

Metric	Count / Value
Total Features Planned	14
Total Features Implemented	14
Core / Must-Have Features	10
Additional / Nice-to-Have Features	4
Features Tested & Verified	14
Overall Implementation Rate	100%